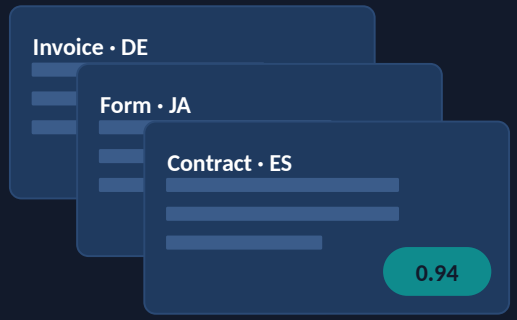
DATANOVA 2026 GLOBAL DATA-THON · TRACK DN-AI-01



Lingua Ledger

Self-checking extraction for multilingual business documents. Every field comes back with a confidence score you can actually trust.

Rishav Naskar · rishavnaskar.r@gmail.com · Round 1 submission, November 2026

Every layout arrives, but trust does not

Official challenge DN-AI-01: extract, classify and structure information from multilingual invoices, contracts and forms with varying layouts and quality, and output structured, validated data with confidence scores.

Multilingual by default

Teams in India, the EU and South-East Asia handle invoices and contracts in 5 to 10 languages, often mixed on one page.

Layouts keep changing

Template OCR breaks on new vendors. LLMs read anything, but they easily make up a VAT number when a scan is blurry.

100% review or errors

Without knowing what is accurate, teams must recheck every field or risk costly ledger errors.

Confidence scores you can actually trust

Approach	Layout robustness	Confidence	Human review
Template OCR	Fragile; breaks on new layouts	Basic word-level OCR confidence only	Humans must check every exception
Prompt-and-pray LLM	Reads almost anything	Self-reported, uncalibrated, and often wrong	All or nothing
Lingua Ledger	Vision LLM + OCR LLM, cross-checked	Calibrated per field: 0.9 means ~90% correct	Only low-confidence fields

Our main metric: at a target field accuracy (like 98%), what share of fields still needs a human? Lower is better, and we measure it directly.

Read twice, validate, score and route

Ingest and classify

Drop in a PDF, scan or photo. We render pages and detect the language and document type (invoice, receipt, form, contract).

Extract twice independently

Path A: a vision model reads the page against a JSON schema. Path B: multilingual OCR text + layout with a text model.

Check with strict rules

Line items must sum, tax math must add up, dates must parse, and IBAN / VAT formats must match. Every rule is clear and explainable.

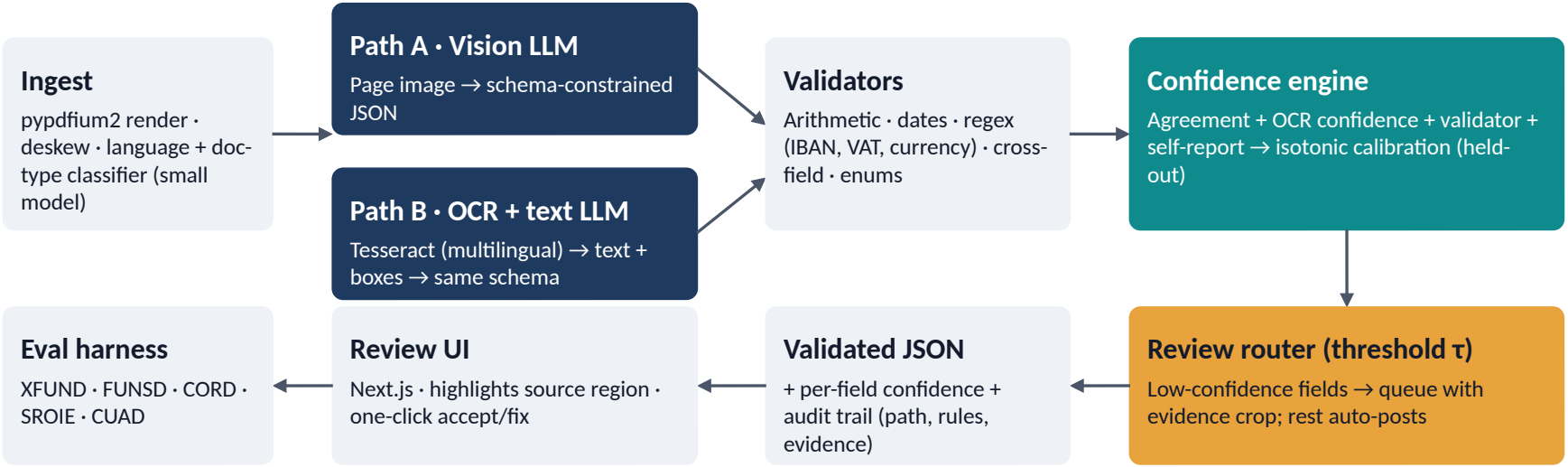
Score confidence per field

We combine reader agreement, OCR confidence, rule results and model estimates, then calibrate them on held-out test data.

Route only doubtful fields

Fields below the threshold go to a review queue with source crops highlighted. Everything else posts automatically with an audit trail.

Two separate readers, one strict judge



Stack: Python 3.13, pydantic schemas, Google Gemini (gemini-3.5-flash-lite for vision, text and classification, structured JSON output; switchable to Claude), Tesseract, scikit-learn (isotonic regression), FastAPI, Next.js on Vercel. Public repo, MIT licence, one command reproduces every metric.

What makes 0.9 mean ninety percent

SIGNALS PER FIELD

Both readers agree
Do the vision reader and the OCR reader return the exact same value?

Clear OCR evidence
Average Tesseract word confidence inside the field box; drops on blur, skew or stamps.

Rule check outcome
Passed, failed or not applicable for every deterministic rule checked against the field.

Raw model estimate
The LLM's own 0 to 1 estimate, used as a small signal, never as the final answer.

Arbiter decision
During disputes, a vision model views a crop around both candidates and picks or re-reads the printed value.

CALIBRATION AND ROUTING

- Signals combine through a small model, then calibrate against field accuracy on held-out data per document type.
- Shown on a reliability diagram with Expected Calibration Error (ECE) before and after calibration.
- A single threshold τ controls routing; operators use the curve to pick their own balance between accuracy and review load.
- Every field carries proof: which path found it, which rules passed, and the exact image crop.

Public benchmarks tested by language and field

Dataset	Document type	Languages	What we measure
XFUND v1.0 (Microsoft)	Forms	ZH, JA, ES, FR, IT, DE, PT	Key-value extraction F1 per language
FUNSD	Forms	EN	Key-value F1; English baseline
CORD v2 (Naver Clova)	Receipts	ID (Latin script)	Field F1; arithmetic validator hit-rate
SROIE (ICDAR 2019)	Receipts / invoices	EN	Exact-match F1 on company, date, address, total
CUAD (Atticus Project)	Contracts	EN	Clause extraction F1 on parties, dates, governing law, termination



Ablation

Vision-only vs OCR-only vs dual-path + validators, same prompts, same schema.



Calibration

Reliability diagram, ECE, Brier score; before vs after isotonic fit, per doc type.



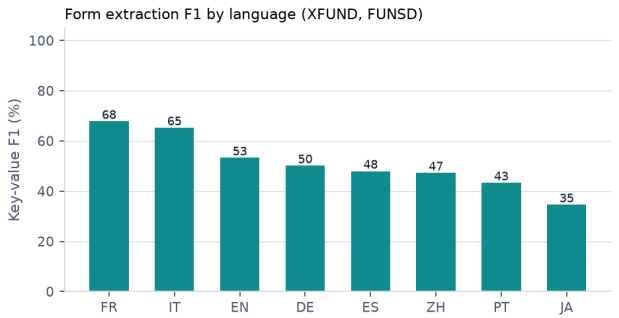
Cost and latency

p50 / p95 seconds per page and dollars per 1,000 documents, by path.

All datasets are public research benchmarks under open licences; no customer data is used. Synthetic multilingual invoices cover the invoice category across the same seven languages.

Results on 198 public benchmark documents

Benchmark	Vision only	OCR + text	Both + validators
XFUND forms (7 langs)	51%	32%	**51%**
FUNSD forms (EN)	54%	39%	**53%**
CORD receipts (ID)	95%	52%	**95%**
SROIE receipts (EN)	95%	89%	**95%**
Synthetic invoices (7 langs)	99%	96%	**99%**
CUAD contracts (EN)	61%	60%	**60%**



92%

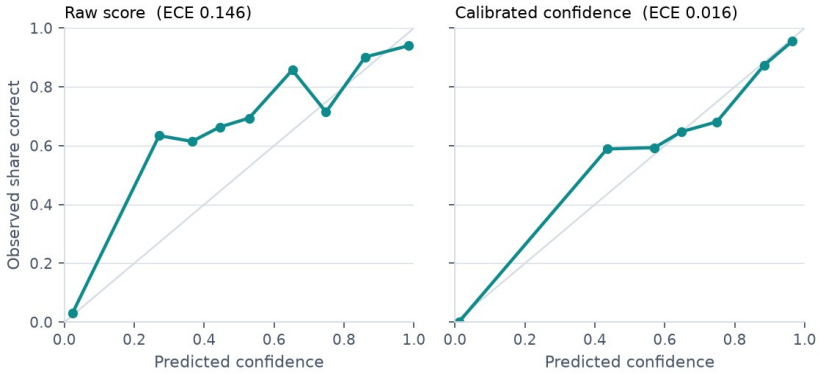
document type detected correctly

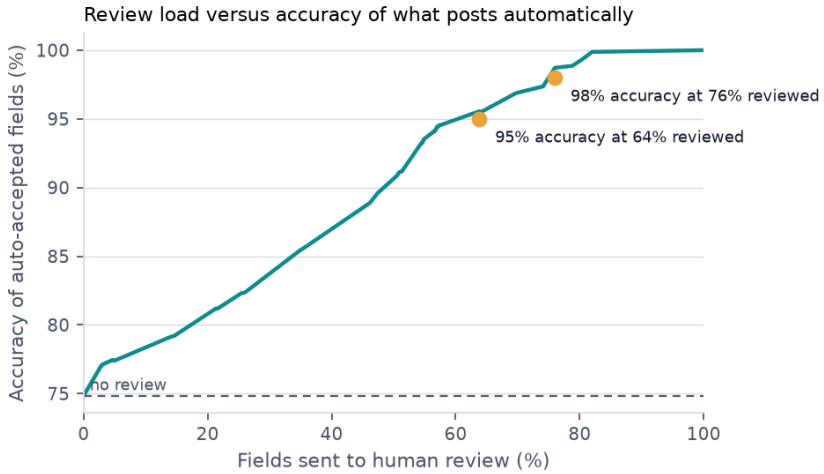
89%

language detected correctly

Field F1 scored against published ground truth. Forms: key-value pairs. Receipts and invoices: every labeled field and line item. Contracts: parties, dates, governing law. Same prompts and schema throughout.

Honest confidence creates a clear review budget





0.146 → 0.016

expected calibration error, raw score → calibrated (lower is better)

75%

fields correct with no human review at all

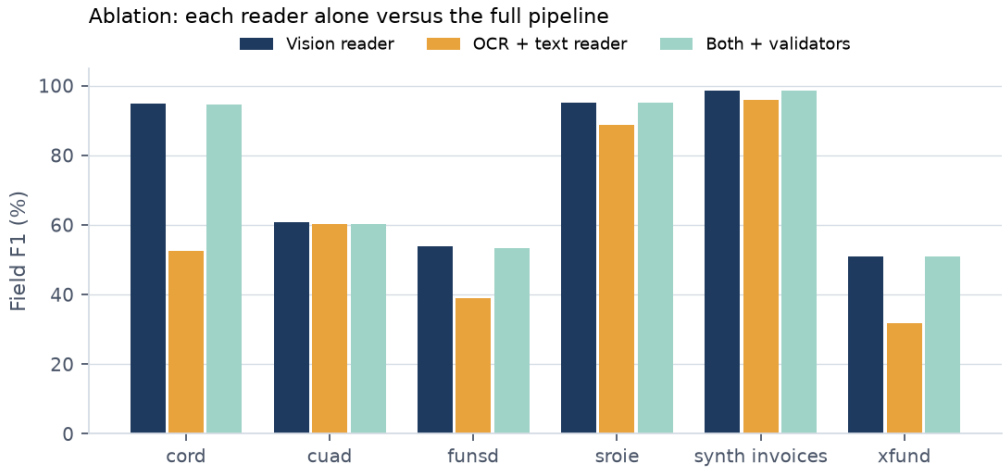
76%

of fields need a human to reach 98% accuracy on what posts automatically

64%

of fields need a human to reach 95% accuracy

What it costs to run



Tokens per document: about 3825 in, 1595 out across both readers. Review effort scales with low-confidence fields, not document volume.

11.8 s

median model reading time per document, both readers in parallel, Free Tier rate limits included

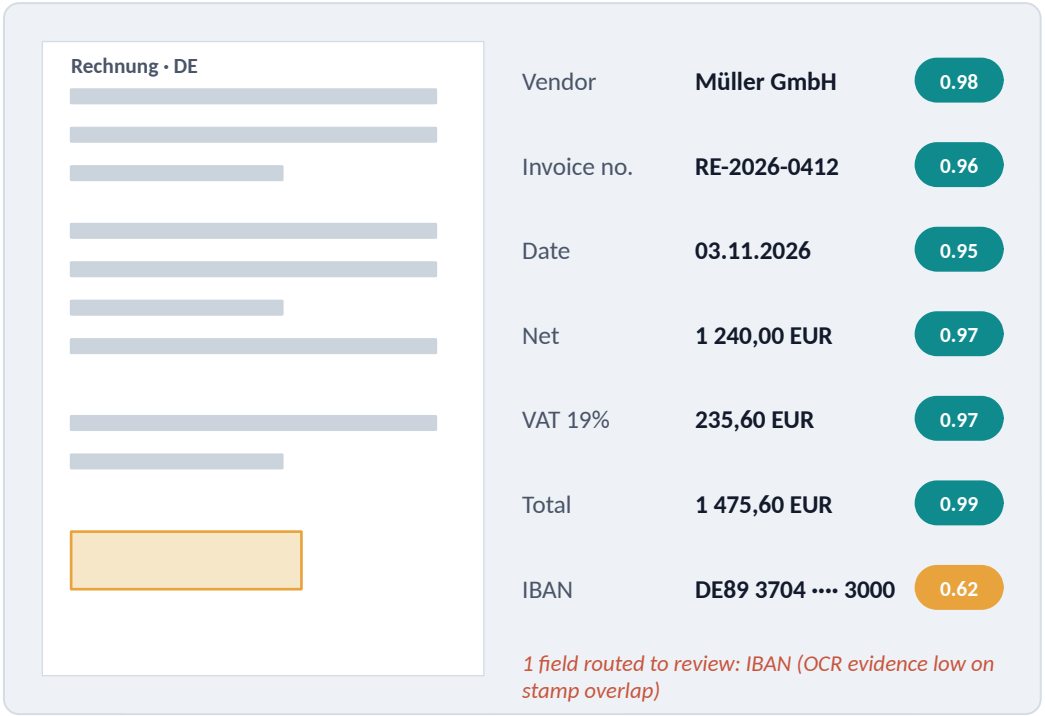
21.2 s

median end to end, with OCR, classification and throttling waits

$5.14

model cost per 1,000 documents at list prices

Demo: drop a file, watch it self-verify



- Upload any PDF or photo; sample files in seven languages are ready to test.
- Side-by-side view showing what each reader saw and where they disagreed.
- Validator panel: see which rules ran and which passed in plain words.
- Threshold slider: move τ and watch review workload and expected accuracy update live.
- Export verified JSON or CSV with a complete audit trail attached.

Fewer manual reviews, zero silent errors

Who benefits

Finance, accounts-payable, procurement and legal teams, plus BPOs working across India, the EU, South-East Asia and LATAM.

Cost savings

Review work scales with low-confidence fields, not document count. Document cost is just one or two model calls per page, as shown in our evaluation.

Easy scaling

Stateless pipeline where every step runs in parallel. Supporting new document types just takes a schema and a rule file, not a new model.

Responsible by design

- Humans step in only when confidence is low; nothing posts automatically below τ.
- Every field carries clear evidence, making every decision easy to audit later.
- Documents stay in memory and are never used for training; PII fields can be masked on export.
- Calibration is reported by language and document type to highlight, not hide, weaker areas.
- Built on public benchmarks, public code and verifiable numbers.

Roadmap: pipeline, evidence, then production

1

Round 1 · 10 Nov

Working pipeline with evidence

- Two readers, an arbiter for disputes, validators, calibration, and review routing
- 198 public benchmark documents, 9 languages, and one command produces all numbers
- Live hosted demo with sample documents, anomaly checks, and calibrator review panel
- 237 unit tests released under an MIT licence

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Round 2 · by 25 Nov

Better readers, same pipeline

- Bigger Gemini and Claude readers, one environment variable, comparing cost and accuracy
- Japanese, Chinese forms: layout-aware prompts, key normalisation, strict and lenient scoring
- Three-way match: reconciled purchase orders, invoices, and receipts, flagging any mismatches
- Backed vendor memory with a database instead of a file

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Beyond Round 2

Moving into production

- Open-weights model path for on-prem use
- Connectors for email inbox, SAP, and Tally
- Full table and handwriting extraction support

TEAM AND CONTRIBUTIONS

RN

Rishav Naskar

Software engineer · AI agents, LLM applications and full-stack web · Solo entrant

- Problem framing, architecture and schema design
- Dual-path extraction pipeline, validators and confidence engine
- Evaluation harness, calibration and reporting
- Review UI, hosted demo, documentation and video

Third-party components acknowledged in the repository: XFUND, FUNSD, CORD, SROIE and CUAD datasets; Tesseract OCR; Google Gemini API; pypdfium2; scikit-learn; FastAPI; Next.js. All code released under the MIT licence.

Lingua Ledger · rishavnaskar.r@gmail.com · repo and demo links in the submission form